

NEO-CCSD Lambda Equations

How the Λ equations change when the reference is a product of an electronic and a protonic determinant: five multiplier blocks, the energy derivatives that drive them, the reversal of every cross coupling under transposition, and the mixed density that the gradient needs.

Written for a graduate student who has read the gradient notes

Spin orbitals, high-spin fermionic protons, NEO Hartree–Fock reference

NOTATION

$i, j, m / a, b, e$	occupied / virtual electronic spin orbitals; $I, J, M / A, B, E$ their protonic counterparts.
μ, ν	labels running over all five amplitude blocks: $e1 = (i, a)$, $e2 = (ij, ab)$, $p1 = (I, A)$, $p2 = (IJ, AB)$, $ep = (iI, aA)$.
$T, \bar{H}$	the NEO-CCSD cluster operator and $e^{-T} H e^T$, as in the amplitude-equation notes.
Λ	the de-excitation operator $\sum_{\mu} \lambda_{\mu} \tau_{\mu}^{\dagger}$, with one block of multipliers per amplitude block.
$J_{\mu\nu}$	the Jacobian $\langle \mu [\bar{H}, \tau_{\nu}] 0 \rangle$, a five-by-five block matrix.
$D_i^a, D_{ij}^{ab}, D_{iI}^{aA}$	orbital-energy denominators, e.g. $D_{iI}^{aA} = \varepsilon_i + \varepsilon_I - \varepsilon_a - \varepsilon_A$.
γ, Γ	response density matrices, $\gamma_{pq} = \langle 0 (1 + \Lambda) e^{-T} a_p^{\dagger} a_q e^T 0 \rangle$ and likewise for the two-particle and protonic cases.

THE WHOLE ANSWER IN THREE LINES

- 1 The Λ equations keep their form, $\langle 0|(1 + \Lambda)(\bar{H} - E)|\nu\rangle = 0$, now projected onto five classes of determinant, and Λ gains a mixed block λ_{iI}^{aA} that de-excites one electron and one proton.
- 2 Each block is driven by the derivative of the energy with respect to its amplitudes: the dressed Fock matrices for the singles, the integrals for the doubles, and $-(ia|IA)$ for the mixed block, so that every block of λ equals the corresponding block of t to first order.
- 3 Because the equations use the transpose of the Jacobian, every cross coupling runs backwards relative to the amplitude equations: where protonic amplitudes fed an electronic equation, electronic multipliers now feed the protonic one.

1 Five multiplier blocks, one linear system

The gradient notes build the Lagrangian $\mathcal{L} = \langle 0|(1 + \Lambda)\bar{H}|0\rangle$ and obtain the Λ equations from its stationarity with respect to the amplitudes. Nothing in that derivation depends on the number of particle types, so for NEO-CCSD the de-excitation operator simply acquires one block per amplitude block,

$$\Lambda = \Lambda_1^e + \Lambda_2^e + \Lambda_1^p + \Lambda_2^p + \Lambda_{11}^{ep}, \quad \Lambda_{11}^{ep} = \sum_{iIaA} \lambda_{iI}^{aA} a_i^\dagger a_a b_I^\dagger b_A, \quad (1)$$

and the equations are projected onto the same five classes of determinant as the amplitude equations:

KEY RESULT · THE NEO-CCSD Λ EQUATIONS

$$\langle 0|(1 + \Lambda)(\bar{H} - E)|\nu\rangle = 0 \quad \text{for } \nu \in \{\Phi_i^a, \Phi_{ij}^{ab}, \Phi_I^A, \Phi_{IJ}^{AB}, \Phi_{iI}^{aA}\}, \quad (2)$$

or in matrix form, with the five-by-five block Jacobian of the amplitude-equation notes,

$$\mathbf{J}^\top \boldsymbol{\lambda} = -\frac{\partial E}{\partial \mathbf{t}}. \quad (3)$$

The bra $\langle 0|(1 + \Lambda)$ is the left eigenvector of $\bar{H}$ in the five-block space, with eigenvalue E , exactly as before. With one quantum proton the protonic doubles block is absent and four blocks remain. The system is linear in λ , perturbation-independent, and solved once; everything the gradient notes say about the $2n+2$ rule and the cost of one more coupled cluster calculation carries over unchanged.

2 The right-hand sides

The inhomogeneous term of each equation is the derivative of the NEO-CCSD energy with respect to the corresponding amplitude, which by the energy formula of the NEO coefficients notes is

KEY RESULT · THE ENERGY DERIVATIVES

$$\begin{aligned} \frac{\partial E}{\partial t_i^a} &= f_{ia}^e + \sum_{jb} \langle ij||ab \rangle t_j^b - \sum_{IA} (ia|IA) t_I^A, & \frac{\partial E}{\partial t_{ij}^{ab}} &= \langle ij||ab \rangle, \\ \frac{\partial E}{\partial t_I^A} &= f_{IA}^p + \sum_{JB} \langle IJ||AB \rangle t_J^B - \sum_{ia} (ia|IA) t_i^a, & \frac{\partial E}{\partial t_{IJ}^{AB}} &= \langle IJ||AB \rangle, \\ \frac{\partial E}{\partial t_{iI}^{aA}} &= -(ia|IA). \end{aligned} \tag{4}$$

In the left-eigenvector form (2) the same quantities appear as $\langle 0|\bar{H}|\nu \rangle$: for the doubles and the mixed double no amplitude can bring the determinant back to the reference through H , so the matrix element is the bare integral, while for the singles the result is the dressed Fock matrix of the amplitude-equation notes. Three consequences follow.

- **Every block of λ equals the corresponding block of t to first order.** The diagonal of the Jacobian is $-D$ for each block, so the leading solution of (3) is $\lambda_{ij}^{ab} \approx \langle ij||ab \rangle / D_{ij}^{ab}$ and $\lambda_{iI}^{aA} \approx -(ia|IA) / D_{iI}^{aA}$, which are the first-order amplitudes. The mixed multipliers are driven directly by the mixed integral, with coefficient 1.
- **The protonic singles multipliers have a driving term.** With a NEO-HF reference the protonic singles amplitudes have none, since $f_{IA}^p = 0$ and their equation is fed only by couplings. Their multipliers, by contrast, are driven by $-\sum_{ia} (ia|IA) t_i^a$, the derivative of the product term in the energy, and by $\sum \langle IJ||AB \rangle t_J^B$ when there are several protons.
- **The singles right-hand sides are dressed Fock matrices.** $\langle 0|\bar{H}|\Phi_i^a \rangle$ is the intermediate $\mathcal{F}_{ia}$ of the single-component working equations with the proton's correction $-\sum (ia|IA) t_I^A$ added, which is the same dressing that appears throughout the amplitude equations.

3 The transpose reverses every coupling

Equation (3) is the transpose of the linearized amplitude equations. A term “coefficient times t_ν in the equation for μ ” becomes “the same coefficient times λ_μ in the Λ equation for ν ”. So the block structure of the Λ equations is read off the term-class table of the amplitude-equation notes by swapping rows and columns. The amplitude equation for a block μ contains t_ν whenever the table lists a product with T_ν in that row, which gives the following pattern of nonzero Jacobian blocks:

AMPLITUDE EQUATION FOR	E1	E2	P1	P2	EP
e1 · electronic singles	•	•	•	–	•
e2 · electronic doubles	•	•	•	–	•
p1 · protonic singles	•	–	•	•	•
p2 · protonic doubles	•	–	•	•	•
ep · mixed doubles	•	•	•	•	•

The four blanks are the only zero blocks: an electronic equation never contains protonic doubles amplitudes, and a protonic equation never contains electronic doubles amplitudes, because the electron–proton attraction has only two lines of each species to spend. Reading the table down a column instead of across a row gives the Λ equations:

- **The electronic singles Λ equation and the mixed Λ equation receive contributions from all five multiplier blocks**, because their columns are full.
- **The electronic doubles Λ equation receives nothing from the protonic multipliers λ_1^p and λ_2^p** , only from λ_1^e , λ_2^e and λ^{ep} . In the amplitude equations the electronic doubles were fed by the protonic singles; after transposition that coupling has moved to the protonic singles Λ equation, where it is fed by λ_2^e .
- **Mirror statements hold for the protonic blocks.**

The direction of every physical coupling therefore flips. In the amplitude equations the electronic doubles drive the mixed amplitude through the term $-\sum_{me}(me|IA)_{im}^{tae}$; in the Λ equations the mixed multipliers drive the electronic doubles multipliers through the same integral. In the amplitude equations the protonic singles dress the electronic Fock matrix; in the Λ equations the electronic singles multipliers appear in the protonic singles equation with the

same mixed integral. This is nothing deeper than the definition of a transpose, but it is the thing most likely to be coded backwards.

4 The mixed Λ equation

Collecting, for the mixed block, the transposes of the linear cross terms displayed in the amplitude-equation notes, with canonical orbitals and the diagonal Fock terms on the left,

MIXED MULTIPLIERS: DRIVING TERM AND LINEAR TERMS

$$\begin{aligned}
 \lambda_{iI}^{aA} D_{iI}^{aA} = & - (ia|IA) \\
 & - \sum_e \lambda_i^e (ea|IA) + \sum_m \lambda_m^a (mi|IA) - \sum_E \lambda_I^E (ia|EA) + \sum_M \lambda_M^A (ia|MI) \\
 & - \sum_{jb} \lambda_{ij}^{ab} (jb|IA) - \sum_{JB} \lambda_{IJ}^{AB} (ia|JB) \\
 & - \sum_{me} \lambda_{mI}^{eA} \langle ie||ma \rangle - \sum_{ME} \lambda_{iM}^{aE} \langle IE||MA \rangle + (\text{terms with amplitudes}).
 \end{aligned} \tag{5}$$

Line by line: the driving term from (4); the singles multipliers of each species entering through the mixed integral, the transposes of the terms by which the mixed amplitude equation was fed by singles; the doubles multipliers of each species, the transposes of the terms by which the electronic and protonic doubles equations were fed by the mixed amplitude; and the ring terms, transposes of the ring terms of the mixed amplitude equation, with the proton or the electron as spectator. The shape mirrors the mixed amplitude equation term for term, but each coupling is the transpose of the corresponding one there, not the same one.

The terms with amplitudes come from the dependence of the Jacobian on t . The amplitude equations are at most quartic in t , so the Λ equations are linear in λ and at most cubic in t , as in the single-component case, and the products that can appear follow from the term-class table by the same transposition. All of the terms in which a singles amplitude multiplies an integral can be absorbed by dressing the integrals with the singles of each species, so a T_1 -transformed implementation of the Λ equations needs only the doubles and mixed amplitudes explicitly.

5 Cross terms in the other Λ equations

The same transposition gives the leading cross terms elsewhere. In the electronic singles Λ equation,

ELECTRONIC SINGLES MULTIPLIERS: LEADING CROSS TERMS

$$-\sum_{IA} \lambda_I^A (ia|IA) - \sum_{eIA} \lambda_{iI}^{eA} (ea|IA) + \sum_{mIA} \lambda_{mI}^{aA} (im|IA), \quad (6)$$

the first from the protonic singles amplitude equation, the other two from the singles terms of the mixed amplitude equation. In the electronic doubles Λ equation the only cross term at this order is the transpose of the doubles-to-mixed coupling,

ELECTRONIC DOUBLES MULTIPLIERS: LEADING CROSS TERM

$$-P(ij)P(ab) \sum_{IA} \lambda_{iI}^{aA} (jb|IA), \quad (7)$$

and the protonic singles and doubles Λ equations contain the mirror images of (6) and (7) with the species swapped. The permutation operators in (7) are there because the electronic doubles are antisymmetric; (5) has none, because the two de-excited particles in the mixed block are distinguishable.

6 Solving the equations

Nothing changes in the method. Each block is updated from its own denominator,

$$\lambda_{iI}^{aA} \leftarrow \frac{-(ia|IA) + R_{iI}^{aA}(\lambda; t)}{D_{iI}^{aA}}, \quad (8)$$

alongside the other four, with DIIS over the concatenated residual of all blocks. The converged amplitudes are fixed input, and because the system is linear in λ it typically converges in fewer iterations than the amplitude equations from the starting guess $\lambda = t$. The cost is that of one more set of amplitude iterations, dominated as before by the electronic particle-particle ladder; the mixed block is small and its contractions scale no worse than $\sigma^2 v^2$ times protonic dimensions. Converge the multipliers as tightly as the amplitudes: the gradient is first order in the residual of both, as the gradient notes explain.

7 Response densities and the gradient

The multipliers exist to build densities. With the operator ordering of the gradient notes, the NEO-CCSD response densities are the electronic and protonic one- and two-particle densities and a new mixed two-particle density,

$$\begin{aligned}\gamma_{pq}^e &= \langle 0 | (1 + \Lambda) e^{-T} a_p^\dagger a_q e^T | 0 \rangle, & \gamma_{PQ}^p &= \langle 0 | (1 + \Lambda) e^{-T} b_P^\dagger b_Q e^T | 0 \rangle, \\ \Gamma_{pqPQ}^{\text{ep}} &= \langle 0 | (1 + \Lambda) e^{-T} a_p^\dagger a_q b_P^\dagger b_Q e^T | 0 \rangle.\end{aligned}\tag{9}$$

Their simplest blocks show the mixed multipliers at work. Where $a_a^\dagger a_i$ or $a_a^\dagger a_i b_A^\dagger b_I$ is an excitation operator it commutes with T , so the similarity transform does nothing and the bra picks out one multiplier:

$$\gamma_{ai}^e = \lambda_i^a, \quad \gamma_{AI}^p = \lambda_I^A, \quad \Gamma_{aiAI}^{\text{ep}} = \lambda_{iI}^{aA}.\tag{10}$$

The occupied and virtual blocks of the one-particle densities contain products of amplitudes and multipliers, and the mixed pair now contributes to both species. For the protonic occupied block, one commutator with each part of T gives

$$\gamma_{IJ}^p = \delta_{IJ} - \sum_E t_I^E \lambda_J^E - \sum_{iaE} t_{iI}^{aE} \lambda_{iJ}^{aE} - \frac{1}{2} \sum_{MEF} t_{IM}^{EF} \lambda_{JM}^{EF},\tag{11}$$

with the mixed term carrying no $\frac{1}{2}$ because no antisymmetric index pair is summed, and the electronic occupied block picks up the analogous $-\sum_{aIA} t_{iI}^{aA} \lambda_{jI}^{aA}$. The proton density, which is the observable that multicomponent methods are judged by, is γ^p back-transformed to the protonic basis, so the mixed multipliers enter it directly through (11).

In the gradient, each density contracts with the derivative of its own integrals. Writing μ, ν for electronic and Λ, Σ for protonic basis functions,

KEY RESULT · THE NEO-CCSD GRADIENT

$$\begin{aligned}\frac{dE}{dx} &= \sum_{\mu\nu} D_{\mu\nu}^e h_{\mu\nu}^{e,x} + \frac{1}{2} \sum \Gamma_{\mu\nu\lambda\sigma}^e (\mu\nu|\lambda\sigma)^x - \sum_{\mu\nu} W_{\mu\nu}^e S_{\mu\nu}^{e,x} \\ &+ \sum_{\Lambda\Sigma} D_{\Lambda\Sigma}^p h_{\Lambda\Sigma}^{p,x} + \frac{1}{2} \sum \Gamma_{\Lambda\Sigma\Pi\Theta}^p (\Lambda\Sigma|\Pi\Theta)^x - \sum_{\Lambda\Sigma} W_{\Lambda\Sigma}^p S_{\Lambda\Sigma}^{p,x} \\ &- \sum_{\mu\nu\Lambda\Sigma} \Gamma_{\mu\nu\Lambda\Sigma}^{\text{ep}} (\mu\nu|\Lambda\Sigma)^x + \frac{dV_{nn}}{dx}.\end{aligned}\tag{12}$$

Here D^e and D^p are the relaxed densities, each carrying its own Z-vector contribution, and the two multiplier vectors $\mathbf{z}^e$ and $\mathbf{z}^p$ come from one coupled linear system in which the NEO-HF orbital Hessian links the species through the electron–proton attraction. The mixed density contracts with the derivative of the mixed Coulomb integrals, which depends on the centres of the electronic basis functions and of the protonic ones. For a gradient with respect to a classical nucleus the electronic functions on that nucleus move with it; whether a protonic basis centre also moves is a modelling choice that has to be stated, as the Thouless notes remark. The energy-weighted densities and the overlap-derivative terms appear once per species, since each set of orbitals is orthonormal on its own.

8 Further reading

1. J. Gauss, J. F. Stanton and R. J. Bartlett, *J. Chem. Phys.* **95**, 2623 (1991); E. A. Salter, G. W. Trucks and R. J. Bartlett, *J. Chem. Phys.* **90**, 1752 (1989). The single-component Λ equations and response densities that each species block reproduces.
2. H. Koch, H. J. Aa. Jensen, P. Jørgensen, T. Helgaker, G. E. Scuseria and H. F. Schaefer III, *J. Chem. Phys.* **92**, 4924 (1990). The Lagrangian formulation of CC derivatives, which transfers to any number of particle types.
3. F. Pavošević, T. Culpitt and S. Hammes-Schiffer, *J. Chem. Theory Comput.* **15**, 338 (2019). NEO-CCSD, with the amplitude equations whose transposes these notes describe.
4. F. Pavošević and S. Hammes-Schiffer, *J. Chem. Phys.* **151**, 074104 (2019). Proton densities from multicomponent CCSD, which require the protonic response density of Section 7.
5. F. Pavošević, T. Culpitt and S. Hammes-Schiffer, *Chem. Rev.* **120**, 4222 (2020). The NEO framework and its correlated methods.

Equations are numbered consecutively through the notes. Equations (5) to (7) show leading terms with the sign conventions of the refresher and are obtained by transposing the cross terms of the amplitude-equation notes; the complete equations should be derived from, or checked against, the amplitude equations of Pavošević, Culpitt and Hammes-Schiffer (2019) in the conventions of a particular code.